PI

CUSTOMER ANALYTICS & CLV · CASE STUDY

# Customer Intelligence: Retention Engine & CLV Suite

BI Scientist • AI/ML Engineer • Python Developer

Calibrated Telco churn prediction (PR-AUC 0.841) generating Top-50 weekly priority intervention lists covering $60,669/year at risk.

RETENTION INEFFICIENCY

# Why Blanket Retention Campaigns Waste Budget

Companies waste millions offering discounts to customers who were never going to leave—while ignoring high-value accounts quietly preparing to churn.

## Uncalibrated Probabilities

- Raw classifier scores do not reflect real churn probabilities.
- Teams cannot calculate expected revenue loss ($P( ext{churn}) imes ext{CLV}$).
- Budget is distributed uniformly instead of by financial risk.

## Contact Budget Constraints

- Account management teams can only contact 10%–20% of subscribers per month.
- Need models optimized for Recall@k% contact budget.
- Standard accuracy fails to prioritize the highest-risk revenue.

## Static vs Transactional CLV

- Historical spend ignores recent purchasing frequency and recency.
- Must combine churn probabilities with forward-looking BG/NBD CLV.
- Target retention offers only where ROI is positive.

RETENTION PIPELINE

# Calibrated GBDT, RFM Segmentation & Action Lists

Translating statistical predictions into immediate operational dollar savings.

## Calibrated Churn Engine

- HistGradientBoosting with isotonic calibration.
- PR-AUC = 0.841, ROC-AUC = 0.933.
- Captures 80.5% of the theoretical ceiling at a 20% budget.

## Transactional CLV Suite

- RFM customer scoring & segmentation.
- 180-day holdout backtest within +1.19% of actuals.
- Quantifies 12-month expected customer value.

## Weekly Action List

- Ranks subscribers by expected dollar loss.
- Generates Top-50 weekly priority call list.
- Flags $60,669.37 in annual contract value at risk.

ACCOUNT MANAGEMENT UI

# Customer Intelligence Retention Dashboard

Streamlit application generating weekly priority call lists and customer risk breakdowns.

- **Top-50 Priority Intervention Queue:** Directly displays customer ID, contract type, churn probability, and exact 12-month revenue at risk.
- **Segment Scatter & RFM Analysis:** Visualizes Champions, At-Risk, and Hibernating customer cohorts.
- **Decision Support:** Recommends specific retention interventions (pricing adjustments vs service check-ins).

Churn Modeling | CLV | Streamlit | Telco Data




github.com/ikebude/ai-ml-data-engineering/tree/main/machine-learning/customer-retention

QUANTIFIED IMPACT

# Verified Performance on 7,043 Subscribers

Exact figures from reproducible `reports/metrics.json` runs.

| PR-AUC | ROC-AUC | Recall @ 20% Budget | Revenue at Risk |
|---|---|---|---|
| 0.841 | 0.933 | 0.607 | $60,669 |
| 95% CI: 0.807–0.871 | Baseline LR was 0.929 | 80.5% of theoretical ceiling | In the top-50 call list |

## Precision at High Recall

At a 20% outreach budget, precision is 80.5% (4 out of 5 contacted were true churners).

Eliminates wasted discount spend on non-churning subscribers.

Bootstrap 95% confidence intervals confirm statistical robustness.

## Forward-Looking CLV Accuracy

180-day holdout CLV validation predicted total revenue within +1.19% of actual receipts.

Fully automated in CI with 2/2 Pytest unit tests passing.

Ready-to-use CSV exports for CRM sales team integration.

OPEN-SOURCE & REPRODUCIBLE

# Turn churn probabilities into prioritized revenue protection.

Customer Intelligence demonstrates how calibrated machine learning directly protects the bottom line.

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**Code:** github.com/ikebude/ai-ml-data-engineering